

Assignment 6: Market Research Report

Virtual Reality in Medical Education

Amanda M. Clark, Cooper Sorich, Marisol Villalobos

Morgridge College of Education, University of Denver

LIS 4011: Information Access & Retrieval

Dr. Shimelis Assefa

June 6, 2024

Executive Summary (Marisol Villalobos)

Virtual Reality (VR) is a computer-generated simulation of an environment where interaction with the software is often a feature. This market research report identifies key strengths in utilizing VR as a learning tool in the classroom for medical education. Future trajectories are explored and findings indicate the market size for this technology is expected to grow exponentially. Key findings include increased knowledge acquisition from simulations, improved confidence and skills in training, and opportunities for more frequent training in medical institution programs. Common concerns include budget costs, technology support, ethical considerations, and usability. In reviewing future trajectories, key findings, and concerns, some recommendations include evaluating the product through an ethical lens, investing in dominant players, and taking into consideration the rapid change and growth of this market to inform decisions about entering this market.

Table of Contents

Title Page.....	1
Executive Summary.....	2
Table of Contents.....	3
Introduction.....	4
Methodology.....	5
Industry Overview.....	6
Market Analysis.....	10
Competitive Analysis.....	12
Consumer Analysis.....	14
Product Analysis.....	16
Marketing Sales and Strategies.....	19
Financial Analysis.....	20
Recommendations.....	21
Conclusion.....	22
Appendices.....	25
References.....	29

Introduction (Amanda Clark)

Scholars believe that medical education began as early as the 5th century BC with the ancient Greeks introducing concepts of how to practice medicine, closely followed by Ayurvedic techniques in India and the development of instruction in Chinese medicine. As these practices evolved, new technologies accompanied them. During the Middle Ages (roughly 476 A.D. to 1450) universities in Italy and across Europe began training physicians formally. Throughout the medical renaissance that followed, many technical innovations appeared such as the mapping of blood circulation, publications on understanding anatomy, and new surgical techniques (Toledo-Pereyra, 2015). “Western medicine” was introduced shortly thereafter (“Timeline of medical education,” 2024). It wasn’t until 1765 that formal American medical education began at the College of Philadelphia, currently known as the University of Pennsylvania, where models of diseased body parts made from wax were used to teach medical students (Fee, 2015).

Throughout the cultural and geographical landscapes, the number of medical technologies introduced are too vast to name. These technologies have evolved for millennia but one of the more modern advancements is the use of virtual reality (VR) in medical education. VR is being used to analyze and problem solve; communicate, collaborate, and grow other soft skills; internalize practical knowledge and procedures; memorize factual knowledge (ex: anatomy of the human body); learn languages; and adjust behaviors (such as those that contribute to positive medical outcomes).

The virtual reality market in medical education is an industry worth watching. Simulation has long been a strong pedagogical tenet in medical studies, and VR serves as one method of simulation that continues to produce favorable results in the outcomes of defined learning objectives in many medical fields. When utilizing VR, medical students put on headsets and can

become immersed in replications of realistic clinical experiences via educational software programs. This report focuses on the virtual reality industry in relation to its use in medical education and will not cover the use of VR in other educational fields or business sectors. VR in medical education is growing in popularity and scholars assessing its future emphasize that it is an inexpensive, dependable, low-risk way for students to acquire valuable skills. Additionally, VR technology can be extended to more remote learners and practicing physicians (Saleh, 2024, p.8). In this report, the authors will comment on industry forces, questions about the VR market with regard to its place in medical education, and the prospects of continuing to utilize VR in medical education.

Methodology (Marisol Villalobos)

The research primarily utilized business and medical journals with supporting information from information technology journals. First, searches were completed using MarketResearch.com Academic database, which offers insight into market research reports covering a variety of disciplines, to get an idea of the dialogue surrounding the uses of Virtual Reality (VR) in medical education. Results that populated from here allowed the team to narrow down the research topic and led us to complete more refined searches in databases such as Academic Search Complete, ProQuest, Taylor & Francis Journals, and Fortune Business Insights.

For the Industry Overview, Market Analysis, and Competitive Analysis, primary data in the form of text, charts, and graphs was retrieved from websites publishing information regarding industry analytics. The Introduction, supporting material, and secondary data was gathered from library databases by utilizing Google Scholar to see what full text documents were available through the University of Denver. Query phrases included: "virtual reality", "vr" OR

“augmented reality”, “medical education”, “medical school”, “medical students”, “medical curriculum”, “medical student education”, “clinical education”, “swot analysis”, “market size”, “forecast OR growth”, “evolution”, “competitors”, “ethic*”, “geography”, and “key players.”

Specific databases used to complete market share and current/future forecast data included Oman Medical Journal, Fortune Business Insights, and SICCODE.com.

For the Consumer Analysis, systematic reviews were used to inform the report with user attitudes, concerns, and benefits of VR in medical education. Queries included “virtual reality AND medical training AND challenges” or “virtual reality AND medical training AND benefits.” The Marketing and Sales Strategies section focused on analyzing Meta’s sites directly and included their social media posts referencing VR in medical education. For the financially analysis, alternative web sources were used.

Industry Overview (Amanda Clark)

Several of the industry journal articles cited throughout this research report explain that exponential growth is expected in the medical education sector and that a large part of that growth is attributable to advances in the current and potential uses for VR and AR (augmented reality). “VR simulates real-world objects, events, locations, and interactions in 3D multimedia sensory environments. It has been used successfully in medical education for surgical training, learning anatomy, and advancing drug discovery” (Beverly et al., 2022, p.1)

As mentioned, simulation has a long history with regard to its usefulness in the education of medical students. In the latter half of the 20th century, there were three distinct movements (see Appendix A) that bolstered the argument for the use of simulation in the medical landscape. These developments include the use of manikins to mimic resuscitation, simulating anesthesia on manikins, and a medical education reform movement centered around the acknowledgment that

young doctors often do not feel prepared for clinical practice upon graduation from medical school (Bradley, 2006). Medical students and doctors welcome advances in simulation technologies to help streamline their early training and to strengthen the continuing education of higher professionals. The drivers that support innovation in the use of simulation technologies in medical education are many (see Appendix B). Some of these drivers include patient safety, team-based learning, time restrictions of working professionals, and the failure of older training models to establish preparedness (Bradley, 2006).

More recently, in the first part of the 21st century, both low-technology, like simulated patients (trained actors), and high-technology simulators, like VR and haptic systems, have helped to alleviate the deficiencies defined in the previous paragraph. Some of the trends and developments that have proven successful to some degree in the last five years have included using VR headsets and AR glasses to:

- Visualize diabetes in Appalachia (Beverly, 2022)
- Train professionals on sleeve gastrectomy (Barteit et al., 2021)
- Display instructions on procedural steps of catheter insertion (Barteit et al., 2021)
- Show 360° videos of surgical knot training (Barteit et al., 2021)
- Teach hip arthroplasty (Barteit et al., 2021)
- Simulate suture skills to medical students (Barteit et al., 2021)
- Demonstrate peg transfer (Barteit et al., 2021)
- Teach needle insertion for central venous catheter placement (Barteit et al., 2021)
- Train future surgeons in laparoscopic cholecystectomy (Barteit et al., 2021)
- Test the feasibility of ultrasound procedures (Barteit et al., 2021)
- Teach needle chest compression to nursing students (Barteit et al., 2021)

- Stress test emergency personnel in basic triage for a mass accident event (Barteit et al., 2021)
- Practice ophthalmoscopic examination (Barteit et al., 2021)
- Practice dental implantation (Barteit et al., 2021)
- Replicate hysterectomy training (Barteit et al., 2021)
- Study angiogram models (Barteit et al., 2021)
- Teach urogynecologic surgical anatomy using an interactive holographic module (Barteit et al., 2021)
- Increase understanding of neurodegenerative diseases (Barteit et al., 2021)

Some of the companies and their associated products used in the above-mentioned simulations include the HTC Vive, the Brother AirScouter, the Oculus Rift and Rift CV1, the Microsoft HoloLens, Google's Glass, Samsung's Gear, the ARO, the EyeSI BIO, Sony's HMS T1, and Google's Daydream. In terms of consumer perception and preferences when viewing medical students and medical professionals as the consumers, Barteit et al. (2021) explain that:

- Increases in self-confidence and self-esteem were noted in some instances
- Students and professionals sometimes ranked VR and AR higher than conventional training methods
- HMD (head mounted display) technologies were described as viable alternatives to traditional approaches
- VR has the potential to improve the treatment of cancer on a global scale
- More accurate placement of medical tools was seen in several fields
- Simulation may be a better way to teach certain sets of medical skills

- In spaces where VR was not superior to older methods, it may still be a complementary tool
- Training participants were more enthusiastic about the material
- Proficiency and recall were improved in several instances

When viewing hospitals and universities as the consumers, return on investment is a baseline consideration. A study published in 2018 analyzing the economy of using VR for pain reduction therapy found that money was saved only when use of the VR pain therapy reduced the length of the patient's stay (Delshad et al., 2018). However, return on investment is not the only facet for consideration. Kopel et al. (2021) write about the virtues of encouraging creativity and enjoyment during education and professional skill building:

A recent study by the University of Toronto showed that game-based learning compared with traditional case-based learning improved long-term retention of essential skills on stroke prevention and management. A similar study by the Stanford University Department of Emergency Medicine showed that the 360 Virtual Reality (360 VR) platform for disaster event decision-making was received well by attendings, residents, and medical students and provided a novel method for immersive educational experiences. (p. 823)

Kopel et al. (2021) conclude that students who are offered unique methods of engagement with learning materials perform better at work and on exams, have higher self-esteem, and enjoy greater course satisfaction and better clinical experience. Further, they offer that entertaining experiences help hold the attention of the mind and improve executive function (memory), recollection, and deep learning (p. 823).

The ethical consequences of VR use in medical education can be either favorable or undesirable depending upon the circumstance. Favorable outcomes include increased patient safety reducing medical errors that cause death, enhanced clinical competence, improved opportunity for competency testing, shorter periods of training being needed, and better access to medical education in rural and remote geographies. Undesirable outcomes include:

- Potential hubris caused by over-training in successful completions of VR simulations causing neglect of early decision-making skills that may have ruled out the need for surgery or other procedures in the first place
- Inequitable access due to the financial circumstance of the student and/or the medical program
- Further stratification of economically disadvantaged students
- Encouraging newer clinicians to default to more expensive procedures resulting in higher costs for medical services for conditions that could have been treated using cheaper traditional methods

Therefore, the regulations will need to be developed “to guard against the financial discrimination that will affect not only trainees, but also the institutions that treat the least fortunate populations in our society” (Iserson, 2018, p. 331).

Market Analysis (Amanda Clark)

A thorough market analysis of VR in medical education requires a global view of the virtual reality industry, as well as the medical education industry taken separately and together. According to the industry classification experts at siccode.com, future growth is predicted for the “Computers – Virtual Reality” industry, NAICS code 541512-08, in the United States:

The industry is expected to grow at a rapid pace due to the increasing demand for virtual reality technology in various sectors such as healthcare, education, and entertainment.

The COVID-19 pandemic has also accelerated the adoption of virtual reality technology as it provides an immersive experience without the need for physical contact. The industry is expected to continue to innovate and develop new applications for virtual reality technology, which will further drive growth (NAICS Code 541512-08—Computers-Virtual Reality, 2024, para.7).

Similarly, medical education market size is expected to increase to 185.66 billion USD at a rate of 20.88% between 2023 and 2028 (see Appendix C) (Medical Education Market Size, Trends, Analysis, Forecast 2024-2028, 2024, para. 1).

The virtual reality industry contains business segments outside of the medical and healthcare sectors such as gaming, real estate, and film/entertainment, which contribute to its overall market size. Similarly, the medical education industry contains multiple segments as well. The “blended learning” segment of the medical education industry, which includes in-person and online coursework, is also poised to experience significant growth. A large portion of this growth is attributed to the gamification of learning using technologies such as AR and VR (Medical Education Market Size, Trends, Analysis, Forecast 2024-2028, 2024, para. 4). When VR and medical education, outside of their larger markets (computers and healthcare), are taken together, it becomes clear that these segments are well matched for continued growth.

Researchers at Fortune Business Insights estimate that “the global Virtual Reality (VR) in healthcare market size is projected to grow from \$4.18 billion in 2024 to \$38.46 billion by 2032, at a CAGR of 32.0%” (see Appendix D) (Virtual Reality in Healthcare Market | Revenue Statistics [2032], 2024, para. 1).

Beyond the business segments, this industry is also segmented into its hardware, software, and content components. Hardware leads in market share with the sales of mobile devices, laptops, PC's, and head mounted displays (HMD's). Hardware is followed closely by software given it is also in high demand as healthcare professionals continue to request enhanced training opportunities (see Appendix E) (Virtual Reality in Healthcare Market | Revenue Statistics [2032], 2024, para. 5).

Immersive VR technologies are being used in medical education in North America more so than in other regions. A North American market share of 1.17 billion was recorded in 2023. Research funding in the United States has contributed to this dominance. However, the Asia Pacific, Europe, Middle East/Africa, and South American regions are also expecting expansion in investments in VR in their medical and healthcare sectors (Virtual Reality in Healthcare Market | Revenue Statistics [2032], 2024, para. 6).

Competitive Analysis (Amanda Clark)

Multiple technology companies have developed virtual reality hardware, software, and content that supports medical students and the ongoing education of healthcare professionals. Currently, the companies that offer the most competitive technology in the market are:

- Koninklijke Philips N.V. (Netherlands)
- Samsung Electronics Co. Ltd. (South Korea)
- EON Reality (U.S.)
- CAE Healthcare (U.S.)
- Oculus Rift (Facebook Inc.) (U.S.)
- Medical Realities Ltd. (U.K.)
- XRHealth USA Inc. (U.S.)

- Amelia (formerly Psious VR) (Spain)
- HTC Corporation (Taiwan)
- Sony Corporation (Japan)
- PsyTech VR (U.S.)

Some notable recent competitive products and developments from these and other VR leaders include AI upgrades, integrated medical imaging, virtual therapy for autism, improvements to surgical planning and navigation software, a non-invasive cardiovascular simulation, and expanded funding for Metaverse treatment options. Some of the companies that publish content are trending toward offering “microlearning” or short duration modules to cater to users with time constraints (Virtual Reality in Healthcare Market | Revenue Statistics [2032], 2024, para. 7). VR has also been used to teach empathy to medical students who are treating older adults with age-related conditions (Dyer et al., 2018).

A SWOT analysis of VR in medical education shows that there are about the same number of strengths and opportunities (16) as there are weaknesses and threats (16). However, the potential threats do not seem to be slowing the growth of the market. Rather, many industry experts are working to eliminate threats and address weaknesses with hardware and software improvements, as well as content adjustments. The following chart is a record of all the assessments found in recent literature regarding these four SWOT areas:

Strengths	Opportunities
Simulation preferred over rote methods	User participation in the design process
Repeatable, standardized scenarios	Designing attractive contents
Simple setup	Design of safety parameters
No faculty required	
Learners not confined geographically	
Enjoyable gamified experience	

Cheaper than other simulators, cost-effective	
Reports on effectiveness favorable in many sectors	
Educators able to vary learning methods	
Collaborative learning amplified	
Mistakes without consequences	
Wide range of training applications	
Manageable learning modules	

Weaknesses	Threats
Not suitable to every educational goal	Open source and free education market boom
Reduced face to face communication	High cost of development
Ethical use considerations	High cost of implementation
High implementation cost for smaller schools	Need for specialized skills and expertise
Size and clunkiness of hardware	Safety considerations
Remote technical support	Common side effects
Efficacy of teacher replacement unknown	Fear of expert educator replacement
Learning curve, especially for older adults	Evaluation and validation

(Baniasadi et al., 2020; Medical Education Market Size, Trends, Analysis, Forecast 2024-2028, 2024; NAICS Code 541512-08—Computers-Virtual Reality, 2023; Pottle, 2019; Saleh, 2024)

Consumer Analysis (Marisol Villalobos)

With growing awareness concerning Virtual Reality (VR) and the convenience and comfort it can offer, risk-free structured environments are a major selling point for educators training students in the medical field. This technology allows students to practice without the possibility of harming a patient while offering repeatability of complex clinical cases. According to a systematic review conducted on head mounted displays or HMDs, students perceived their learning environment as motivating and engaging while training (Barteit et al., 2021).

Moreover, VR is shown to improve retention of intricate and complex medical concepts. There are many more benefits to incorporating these tools into medical education. A

meta-analysis of 28 publications was conducted to evaluate not only the effectiveness of VR in medical training but in clinical care as well. Five studies showed improvement in knowledge acquisition, six studies revealed improved skills, and four studies showed improved confidence. While VR was shown to improve students' skills and learning, clinical care saw improvements in rehabilitation and mental health (Dhar et al., 2023).

In addition to the benefits described above, there are some concerns that accompany VR in medical education including reduced face-to-face interactions, usability considerations, and design problems. As previously mentioned, VR in medical training can help students develop empathy, however, it does not provide an environment where students can experience relationship building with human patients. Secondly, because there is no standard for VR in medical education, software and hardware will need to be evaluated using the walk-through method, heuristic evaluation, and/or user experience evaluation to ensure ease of use or discover any problems needing to be resolved. Lastly, sometimes it is not in the user's control to offer feedback before the final design of VR applications, which means there is the possibility of the technology not suiting the training program in a way that is efficient and user-friendly (Baniasadi et al., 2020).

Apart from the benefits of VR in medical education, there are some concerns that influence final purchasing decisions in acquiring VR for medical programs. Some important factors include initial cost, regulatory and ethical considerations, and user support and training.

When implementing VR in a medical training program, there is more to purchase than just the hardware and software. Due to the high-quality technology, high-speed computers, up to

date graphics cards, tracking systems, and current high resolution displays are required (Baniyadi et al., 2020). Some ethical considerations include whether VR training is a good use of resources. Due to initial costs previously mentioned, purchasing VR equipment may not be appropriate if there are equally effective and inexpensive alternatives to use in medical training. Another ethical consideration is the lack of realism and negative transfer in some applications of VR. This means the training environment doesn't prepare the student for a real life situation such as using a controller to perform a high-risk surgery instead of practicing with the actual tools used (Bailey et al., 2023). Lastly, user support and technical training are needed as explored in four studies where users reported technological issues with one study showing 45 percent of students unable to focus on training and instead had to troubleshoot how to use the technology itself (Shorey & Ng, 2021).

Product Analysis (Cooper Sorich)

Virtual and augmented reality have been growing as a prospect and industry for multiple decades. While it may have taken some time for these practices to become effective and efficient, they are currently capable of providing a wide range of benefits to medical education and professionals. Augmented reality allows practitioners to simulate procedures in the real world while providing information and assisting with communication. Ultimately, it serves to elevate the users ability to observe and analyze their situation (Salehahmadi & Hajialiasgari, 2019, p. 137). This is the main difference between AR and VR, as the goal of virtual reality in medical practice is to remove the user from reality in exchange for a heightened sense of immersion within the simulated procedure or event. These methods can be applied to many sections of medical practice such as: cardiac diseases (Bouraghi et al., 2023), laparoscopy (Saleh, 2024),

vascular access training (Bradley, 2006), and many more. However, there is significant room for improvement within this field. These technologies are most frequently used as supplements for students, as the programs vary in terms of the depth of information they can provide (Radianti et al., 2021, p. 131). Furthermore, AR and VR applications are currently struggling to account for the vast amount of medical procedures that have their own specific requirements (Halbig et al., 2022). To surmise, AR and VR technologies have rightfully been gaining attention and appeal over the past few decades, but there are still a few major hurdles they need to overcome if they are to become a core component of medical education. The companies that successfully overcome these barriers will be those that most effectively utilize the various technologies that are synergized to create more realistic virtual and augmented reality.

There are many individual facets that are involved in VR and AR technologies. Augmented reality systems are typically made up of artificial intelligence, software specifically designed for the system, processing units, lenses or screens to display and view the provided content, and sensors to align the software with the physical world (Microsoft, n.d.). This means that most of the manufacturing of these devices is focused on developing adequate programs and artificial intelligence (AI) to analyze the environment and provide relevant information. Ironically, virtual reality is more dependent on physical components to achieve the heightened immersion it is known for. The first and most recognizable aspect of virtual reality is the headset that is used to limit the user's vision exclusively to the rendered environment. Most headsets utilize two high resolution screens to display images to each eye and tracking sensors to mimic the user's movement in the real world (Bardyn et al., 2019). This headset must then be connected to a computer as no existing headset can contain the necessary processing power on its own

(Bardyn et al., 2019). Lastly, just as with augmented reality, software must be developed and installed on the computer according to the needs of the user and situation they are training for. Ultimately, the manufacturing process for these technologies is primarily focused on software development and computational power. While these are preexisting and prevalent industries, the resources that they consume are limited, and there are many competitors vying to acquire a portion of the virtual and augmented reality market.

There are many companies and organizations that are attempting to dominate and innovate this growing market. As is often the case, some of the most prevalent participants are those who were already involved in technology and information. This includes Meta, Sony, and Samsung. Throughout 2023 Meta held the greatest market share of VR and AR headsets ranging from 50 to 60 percent (IDC Research, 2024). Google and Microsoft operate in this industry as well, but currently do not hold a significant share of the market. Other companies offering competitive technologies include Koninklijke Philips, EON Reality, CAE Healthcare, Medical Realities Ltd., XRHelath USA Inc., Amelia, HTC Corporation, and PsyTech VR. While this market is currently dominated by a few companies, it is still intensely competitive. This is because each firm must manufacture a product that is ideally superior to their competition and then have that product approved by the Federal Drug Administration. In order to achieve this goal, the competing companies must develop innovative designs for both the physical and digital components of VR and AR technology.

Technology has been improving at a tremendous pace over the past century, and extended reality is no exception. These improvements will allow new and existing firms to overcome the current limitations of this technology. In addition to the previously mentioned shortcomings, VR

and AR devices would greatly benefit from improvements in their ability to support the clinical workforce (Khan, 2023, p. 270). There are also sectors of medical practice that these technologies are capable of supporting where significant integration has not yet occurred such as physical rehabilitation and mental health services (Khan, 2023, p. 271-272). It is likely only a matter of time until virtual and augmented reality are significantly synthesized into the healthcare industry, but that will only occur once they maximize their utility and efficacy.

Marketing and Sales Strategies (Marisol Villalobos)

This section focuses on how Meta utilizes various marketing and sales strategies to expose their Virtual Reality (VR) technology to the medical community including social media campaigns, developing partnerships with target communities, and showcasing demonstrations to draw in stakeholders.

Meta utilizes multiple social media platforms to share their campaigns for implementing VR in medical education. Their Instagram account has a total of 3.8 million followers and their most recent post alluding to VR in education generated 5,173 likes (Meta, 2022). Meta's YouTube account has a total of 606,000 subscribers and their video introducing the Oculus Rift reached 473,196 views (Bhasin, 2024).

In June of 2022, Meta released an advertisement on YouTube titled, "The Impact Will be Real: Surgical Training" (Meta, 2022) that kicks off a campaign called *Metaverse Impact* which advocates for economic opportunity, privacy, safety and integrity, and equity and inclusion (Meta, 2024). According to Richard Vincent, CEO and Co-Founder of FundamentalVR, the opportunity to train in a safe and accessible setting now with haptic feedback allows medical

teams to gain confidence and surgical proficiency (Meta, 2023). Fundamental Surgery is just one of many partners featured in Meta's campaign for implementing VR in medical training. Johnson & Johnson Institute is another healthcare partner working with Meta and Osso VR software to enhance medical training. According to a case study conducted on Osso VR software and the Oculus for Business Platform, Osso VR-trained students who performed a simulated intramedullary nailing of a tibia scored 233% higher (Oculus Business, 2020). Meta Quest conducted a panel that hosted stakeholders including Dr. David Axelrod from Stanford Virtual Heart and Todd Chang from Children's Hospital Los Angeles. In this demonstration, Chang walked the audience through a VR simulation training of status epilepticus, or life-threatening seizure, which is a rare but high stakes event and is difficult to give students training on due to the urgency of needing to be resolved as soon as possible (Meta Quest, 2017).

Meta had partnered with Children's Hospital Los Angeles (CHLA) to develop the aforementioned VR simulation and due to the initial results, CHLA now requires this training for incoming students. With this success, CHLA and Oculus expanded the low-frequency, high-stakes program to 11 medical institutions. Some hindrances that the Oculus Go headsets solved were not needing dedicated rooms for simulation environments. In addition, medical teams will have more flexibility for training since it will not require trainers to be away from their patient care duties and instead students can still practice methods. Dr. Chang notes brainstorming of other medical scenarios that fall within the parameters of low-frequency, high-stakes such as urgent care or emergency medicine (Oculus VR, 2018).

Financial Analysis (Cooper Sorich)

Understanding the current state of the companies competing in this field is a crucial component of developing a conception of the market as a whole. For simplicity's sake, this analysis will be limited to Meta, Sony, and CAE Healthcare. It should be noted that CAE Healthcare was sold to Madison Industries in October 2023 (Baily, 2023), and that the analysis of this company will be exclusively focused on its healthcare services. The changes in revenue, profit, costs, and stock prices are from the years 2022 and 2023. Meta experienced an increase in revenue (16%), net income (69%), costs (1%) (Meta, 2023, February 1), and stock price (195%) (Meta, 2023, May 31). It is important to identify the fact that the majority of Meta's profits come from the advertisements on its social media platforms. Sony experienced an increase in revenue (35%), net income (6%), costs (18%) (Sony Group Corporation, 2024), and stock price (23.5%) (Yahoo, *Sony Group Corporation, 2024*). The majority of Sony's revenue comes from its sales of electronics. CAE Healthcare experienced increases in revenue (27%), operating income (100%), cost of sales (30%) (CAE Inc., 2023), and stock price (12%) (Yahoo, *CAE Inc, 2024*). Despite being the smallest of the three companies, CAE healthcare managed to achieve the most comparative growth between 2022 and 2023. While this is a small selection of the competition, this data supports evidence of growth within the extended reality industry, and demonstrates how it can be financially beneficial for companies to participate.

Recommendations (Cooper Sorich)

The extended reality market is inherently nuanced, and integration with medical practice introduces even more complications. That being said, there are still a few core tenants that should be considered for any individual seeking to understand or participate in this market. First, one must consider the ethical implications of each kind of technology, and how medical

practitioners are required to abide by strict ethical codes. Any medical device that causes harm to the user or patient will fail to prove effective in this field and would likely fail to receive FDA approval in the first place. Another crucial consideration is the fact that this market is growing and will likely continue to do so for the next few decades . Therefore, if an individual or group is seeking investment opportunities, then they should certainly consider this market. Presently, the safest options would be the companies currently dominating the market such as Meta, Sony, and Samsung. The most pressing recommendation for designers and producers of these technologies is to maximize their efficacy and efficiency.

Between virtual and augmented reality there are unique and shared areas for improvement. Both of these technologies must graduate from their current supplemental role into one in which they are an integral aspect of medical education and practice. In order to achieve this, both forms of technology must alleviate existing complications. One of the most noticeable issues in virtual reality devices is how they are capable of inflicting nausea and other disorienting effects on users. Perhaps the greatest obstacle to this goal is the current inability of these technologies to adapt to the many complexities and specific requirements of different medical procedures, situations, and environments. Additionally, both of these products are heavily reliant on artificial intelligence to provide accurate and relevant information to their use. These recommendations will prove useful to any individual or group seeking to become involved in this market. However, given its complexity and rapidly evolving landscape, industry actors should also remain aware of new findings, reports, and developments.

Conclusion (Cooper Sorich)

This report has demonstrated the primary industry forces within the market for virtual and augmented reality, and provided information on the potential these devices have within medical education and practice. The research presented was primarily sourced from academic journals and databases that focus on healthcare and information technology. Alternative web based sources were used when certain data was unavailable through scholarly publications.

Several key concepts were identified in this report:

- Simulation has a lengthy history within medical science, and extended reality is the future of this field
- This industry is growing rapidly and is expected to continue on this trajectory due to continued technological progress
- The required complexity of the technology is the primary cause of limitations and issues
 - Successful results are not currently consistent, which affects consumers interest in the products
- Many kinds of organizations and companies operate in this market, ranging from smaller groups that specialize in healthcare to the largest technology firms in the world such as Sony.
 - Meta currently holds the greatest market share due to its significant revenue and the fact that it has been investing in virtual reality for many years.
 - The marketing strategies of Meta were analyzed to identify areas of success and domains in need of improvement
- Once the technology is improved these devices will be capable of providing assistance across the entire field of healthcare

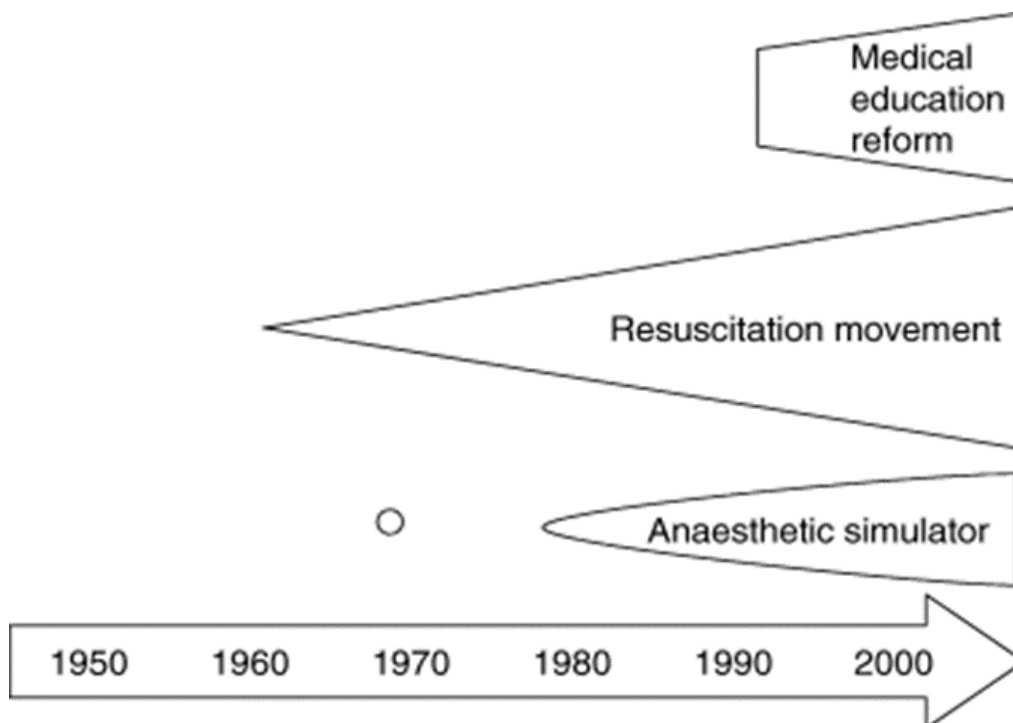
Additionally, certain research topics were determined to be particularly significant to the current state of the industry:

- Improvements in artificial intelligence
- Abatement of negative side effects that occur during and after use
- Strategies and developments that can bolster the engagement provided by these devices for students
- Ethical implications and issues that have or could possibly arise from the production and use of this technology

The information presented provides a macroscopic description of the market for virtual and augmented reality in medical education. Further research is recommended for any individual or group seeking information on specific aspects of the market or products.

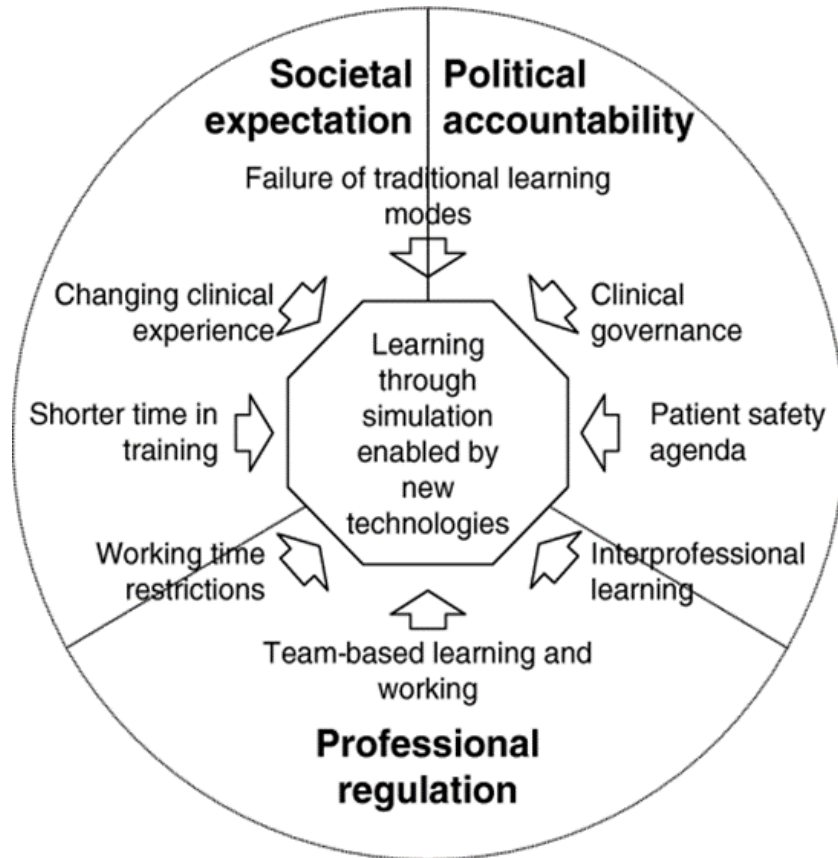
Appendix A

Simulation Movements, Late 20th Century, (Bradley, 2006)



Appendix B

Drivers that Support Simulation Innovation, (Bradley, 2006)



Appendix C

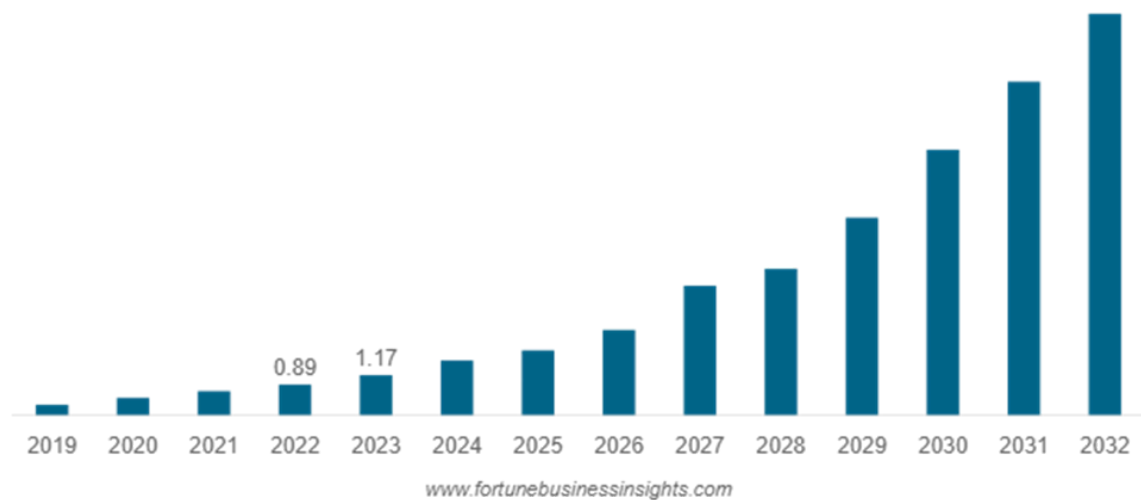
Medical Education Expected Market Growth



Appendix D

VR in Medical Education Market Size

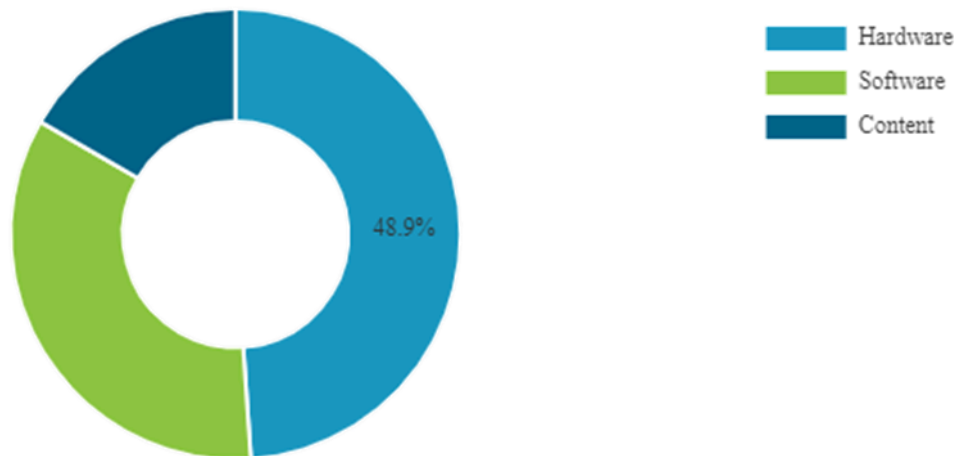
North America Virtual Reality (VR) in Healthcare Market Size, 2019-2032 (USD Billion)



Appendix E

Component Segments

Global Virtual Reality (VR) in Healthcare Market Share, By Component, 2023



www.fortunebusinessinsights.com

References

- Bailey, S., Okuda, Y., & Reiner, C. (2023). *Pros, Cons, and Considerations of Implementing Live Virtual Reality in Medical Education* (pp. 215–229).
<https://doi.org/10.4324/9781003329718-15>
- Baily, L. (2023, October). *CAE Announces Sale of CAE Healthcare to Madison Industries for C\$311 Million*. Healthy Simulation.
<https://www.healthysimulation.com/53211/cae-healthcare-madison-industries/>
- Baniasadi, T., Ayyoubzadeh, S. M., & Mohammadzadeh, N. (2020). Challenges and practical considerations in applying virtual reality in medical education and treatment: Oman medical journal. *Oman Medical Journal*, 35(3), 93–102. <https://doi.org/10.5001/omj.2020.43>
- Bardyn, T., Garret, A., Ruhl, D., Meerovitch, G. (2019). VR Hardware and Software Options. *Virtual Reality in Health Sciences Libraries: A Primer*. University of Washington Libraries. Retrieved from:
<https://uw.manifoldapp.org/read/untitled-a0c48ef4-8048-4562-a27b-80a97e261d51/section/3bd73e2f-10dc-4e8d-89a9-3226050e454b>
- Barteit, S., Lanfermann, L., Bärnighausen, T., Neuhaus, F., & Beiersmann, C. (2021). Augmented, mixed, and virtual reality-based head-mounted devices for medical education: Systematic review. *JMIR Serious Games*, 9(3), e29080. <https://doi.org/10.2196/29080>
- Beverly, E., Rigot, B., Love, C., & Love, M. (2022). Perspectives of 360-degree cinematic virtual reality: Interview study among health care professionals. *JMIR Medical Education*, 8(2), e32657. <https://doi.org/10.2196/32657>

Bhasin, H. (2024, February 16). *Marketing mix of oculus rift and 7ps (updated 2024)*.

Marketing91.

<https://www.marketing91.com/marketing-mix-oculus-rift/#:~:text=Oculus%20Rift's%20entire%20marketing%20field,to%20promote%20its%20VR%20headsets>

Bouraghi, H., Mohammadpour, A., Khodaveisi, T., Ghazisaeedi, M., Saeedi, S., &

Familgarosian, S. (2023). Virtual Reality and Cardiac Diseases: A Systematic Review of Applications and Effects. *Journal of Healthcare Engineering* 2023.

<https://doi.org/10.1155/2023/8171057>

Bradley, P. (2006). The history of simulation in medical education and possible future directions.

Medical Education, 40(3), 254–262. <https://doi.org/10.1111/j.1365-2929.2006.02394.x>

CAE Inc. (2023, May). *Financial Report*.

https://www.cae.com/media/documents/Corporate/Financial_Results/2023/CAE-FY23-FinancialReport_EN.pdf

Delshad, S. D., Almario, C. V., Fuller, G., Luong, D., & Spiegel, B. M. R. (2018). Economic

analysis of implementing virtual reality therapy for pain among hospitalized patients. *NPJ*

Digital Medicine, 1, 22. <https://doi.org/10.1038/s41746-018-0026-4>

Dhar, E., Upadhyay, U., Huang, Y., Uddin, M., Manias, G., Kyriazis, D., Wajid, U., AlShawaf,

H., & Syed Abdul, S. (2023). A scoping review to assess the effects of virtual reality in medical education and clinical care. *Digital health*, 9, 20552076231158022.

<https://doi.org/10.1177/20552076231158022>

- Dyer, E., Swartzlander, B. J., & Gugliucci, M. R. (2018). Using virtual reality in medical education to teach empathy: Journal of the medical library association. *Journal of the Medical Library Association*, 106(4), 498–500. <https://doi.org/10.5195/jmla.2018.518>
- Fee, E. (2015). The first american medical school: The formative years. *The Lancet*, 385(9981), 1940–1941. [https://doi.org/10.1016/S0140-6736\(15\)60950-3](https://doi.org/10.1016/S0140-6736(15)60950-3)
- fortunebusinessinsights.com. (n.d.). *Virtual Reality in Healthcare Market | Revenue Statistics [2032]*. Retrieved May 20, 2024, from <https://www.fortunebusinessinsights.com/industry-reports/virtual-reality-vr-in-healthcare-market-101679>
- Halbig, A., Babu, S., Gatter, S., Latoschik, M., Brukamp, K., & von Mammen, S. (2022). Opportunities and Challenges of Virtual Reality in Healthcare - A Domain Experts Inquiry. *Frontiers of Virtual Reality* 3. <https://doi.org/10.3389/frvir.2022.837616>
- IDC Research. (2024). *AR & VR Headsets Market Insights*. <https://www.idc.com/promo/arvr>
- Iserson, K. V. (2018). Ethics of virtual reality in medical education and licensure. *Cambridge Quarterly of Healthcare Ethics*, 27(2), 326–332. <https://doi.org/10.1017/S0963180117000652>
- Khan, S. (2023). Extended reality for healthcare systems: recent advances in contemporary research. Academic Press. <https://doi.org/10.1016/C2021-0-00530-7>
- Kopel, J., Brower, G., & Culberson, J. W. (2021). Teaching methods fostering enjoyment and creativity in medical education. *Journal of Community Hospital Internal Medicine Perspectives*, 11(6), 821–824. <https://doi.org/10.1080/20009666.2021.1979739>

Meta. (2022, June 15). *The impact will be real: Surgical training*. YouTube.

<https://youtu.be/wcb7kMvDHss?si=93D9dnZHjqs238d>

Meta. (2022, November 4). *Meta on Instagram: "Dr. Morris moved the classroom into the metaverse! she was inspired to change the way students at Morehouse College learn, and now runs a VR teaching program with Meta Quest. the digital twin of the university, built by victoryxr, allows hands-on equitable access to VR that's transforming the future of Education."* Instagram.

https://www.instagram.com/reel/CkjCIAIpCfe/?utm_source=ig_web_copy_link&igsh=MzRlODBiNWFlZA

Meta. (2023, September 7) *For those using Metaverse Technologies, the impact is real*.

<https://about.fb.com/news/2023/09/impact-of-vr-and-ar/>

Meta. (2024). *The impact of the metaverse: Meta*.

https://about.meta.com/metaverse/impact/?utm_source=meta.com&utm_medium=redirect

Meta. (2024, February 1). *Meta Reports Fourth Quarter and Full Year 2023 Results; Initiates Quarterly Dividend*. Meta Investor Relations.

<https://investor.fb.com/investor-news/press-release-details/2024/Meta-Reports-Fourth-Quarter-and-Full-Year-2023-Results-Initiates-Quarterly-Dividend/default.aspx>

Meta. (2024, May 31). *Stock Chart*. Meta Investor Relations. <https://investor.fb.com/stock-info/>

Meta Quest. (2017, October 26). *Oculus connect 4: Improving Healthcare Education Through VR*. YouTube. https://youtu.be/x9YceCGm0kk?si=FO8_H5RwDepOPXgL

Microsoft. (n.d.). *What is augmented reality or AR?* Microsoft.

<https://dynamics.microsoft.com/en-us/mixed-reality/guides/what-is-augmented-reality-ar/>

Oculus Business. (2020). *Virtual reality for surgical training by Johnson & Johnson: Oculus for business*. Virtual Reality for Surgical Training by Johnson & Johnson | Oculus for Business.

<https://business.oculus.com/case-studies/johnson-and-johnson/>

Oculus VR. (2018, August 29). *Immersive education: CHLA and oculus expand VR medical training program to New Institutions*. Meta.

<https://www.meta.com/blog/quest/immersive-education-chla-and-oculus-expand-vr-medical-training-program-to-new-institutions/>

Pottle, J. (2019). Virtual reality and the transformation of medical education. *Future Healthcare Journal*, 6(3), 181–185. <https://doi.org/10.7861/fhj.2019-0036>

Saleh, N. (2024, March 4). *Virtual reality is becoming increasingly useful in medical training—Here's how*. MDLinx.

<https://www.mdlinx.com/article/virtual-reality-is-becoming-increasingly-useful-in-medical-training-here-s-how/2yskJ4fejj2R1bP5Ej8LM5>

Salehahmadi, F. & Hajjaliasgari, F. (2019). Grand Adventure of Augmented Reality in Landscape of Surgery. *World Journal of Plastic Surgery* 8(2). (135-145).

<https://doi.org/10.29252/wjps.8.2.135>

Shorey, S., & Ng, E. D. (2021). The use of virtual reality simulation among nursing students and registered nurses: A systematic review. *Nurse Education Today*, 98, 104662.

<https://doi.org/10.1016/j.nedt.2020.104662>

SICCODE.com. (2023, May 15). *NAICS code 541512-08—Computers-virtual reality*.

<https://siccode.com/extended-naics-code/541512-08/computers-virtual-reality>

Sony Group Corporation. (2024). *Financial Statements and Consolidated Financial Results for the Fiscal Year Ended March 31, 2023 And Outlook for the Fiscal Year Ending March 31, 2024*. https://www.sony.com/en/SonyInfo/IR/library/presen/er/pdf/22q4_sony.pdf

technavio.com, T. (n.d.). *Medical education market size, trends, analysis, forecast 2024-2028*.

Retrieved May 20, 2024, from

<https://www.technavio.com/report/medical-education-market-industry-analysis>

Timeline of medical education. (2024). In *Wikipedia*.

https://timelines.issarice.com/wiki/Timeline_of_medical_education

Toledo-Pereyra, L. H. (2015). Medical renaissance. *Journal of Investigative Surgery*, 28(3), 127–130. <https://doi.org/10.3109/08941939.2015.1054747>

Yahoo. (2024, May 31). *CAE Inc. (CAE)*. Yahoo Finance.

<https://finance.yahoo.com/quote/CAE/history/?period1=1671667200&period2=1703980800>

Yahoo. (2024, May 31). *Sony Group Corporation*. Yahoo Finance.

https://finance.yahoo.com/quote/SONY/history/?guccounter=1&guce_referrer=aHR0cHM6Ly93d3cuZ29vZ2xlLmNvbS8&guce_referrer_sig=AQAAAB5mIgUqvwump2K_VdFwp48KXJHG0fzD0uVQpbR_mRTntumxmlUk4ealsG6LKbOvPCrYPzy92w72CY2CzDaD6t-wdT1mcK5q5VWeKnpuBGPnmmIFqbpRXKuFf3AAOBlkCH2oLU2W5Rencr2s6J9StyPfUu1sy5KsTdP9SMjxXYt&period1=1672444800&period2=1703980800

► Virtual Reality
in Medical Education
Market Research



Amanda Clark, Cooper Sorch, Marisol Villalobos
June 6, 2024



▶ Executive Summary (Marisol)

- ▶ Key Findings:
 - Knowledge acquisition
 - Improved confidence
 - Opportunity for frequent training

- Common Concerns: ◀
 - Budget costs
 - Technology Support
 - Ethical considerations
 - Usability



► Introduction (Amanda)

Two markets merge: Medical Education and Virtual Reality/Augmented Reality (VR/AR)

- Medical education began 5th century BC
- Medical technologies continue to evolve
- VR used to:
 - Analyze/problem solve
 - Communicate, collaborate, grow soft skills
 - Internalize practical knowledge and procedures
 - Memorize factual knowledge
 - Learn languages
 - Adjust behaviors

Most important thing: simulation is a strong pedagogical tenet in medical education and VR enhances our ability to replicate realistic learning experiences.



► Methodology (Marisol)



Research

- MarketResearch.com
- Academic database
- Academic Search Complete
- ProQuest
- Fortune Business Insights



Example Queries

"virtual reality AND medical education"

"virtual reality AND medical training AND benefits"

"virtual reality AND medical training AND challenges"

► Industry Overview (Amanda)



Exponential Growth

Many current and potential uses influence continued growth: surgical training, learning anatomy, and drug discovery are just a few.

3

Evolution

Three distinct movements that bolstered VR in MedEd:

- Resuscitation manikins
- Anesthesia manikins
- MedEd reform began in response to complaints



Trends & Ethics

Deficiencies noted by young doctors are being addressed by VR innovations. Favorable outcomes currently outweigh undesirable outcomes.



► Market Analysis (Amanda)



Market Size

- \$4.18 billion in 2024
- Adoption attributed to pandemic acceleration, blended learning, and gamification



Projected Growth

- \$38.46 billion by 2032
- CAGR of 32%



Segmentation

- MedEd (in-person/online)
- VR (gaming, real-estate, film/entertainment)



Geographical Distribution

- North America leads (\$1.17 billion in 2023)
- Asia Pacific, Europe, Middle East/Africa, South America also expecting expansion

► Competitive Analysis (Amanda)

Strengths

- 1) Simulation preferred over rote methods
- 2) Repeatable, standardized scenarios
- 3) Simple setup
- 4) No faculty required
- 5) Learners not confined geographically
- 6) Enjoyable gamified experience
- 7) Cheaper than other simulators, cost-effective
- 8) Reports on effectiveness favorable in many sectors
- 9) Educators able to vary learning methods
- 10) Collaborative learning amplified
- 11) Mistakes without consequences
- 12) Wide range of training applications
- 13) Manageable learning modules

Weaknesses

- 1) Not suitable to every educational goal
- 2) Reduced face to face communication
- 3) Ethical use considerations
- 4) High implementation cost for smaller schools
- 5) Size and clunkiness of hardware
- 6) Remote technical support
- 7) Efficacy of teacher replacement unknown
- 8) Learning curve, especially for older adults

Opportunities

- 14) User participation in the design process
- 15) Designing attractive contents
- 16) Design of safety parameters

Threats

- 9) Open source and free education market boom
- 10) High cost of development
- 11) High cost of implementation
- 12) Need for specialized skills and expertise
- 13) Safety considerations
- 14) Commonsense effects
- 15) Fear of expert educator replacement
- 16) Evaluation and validation

► Consumer Analysis (Marisol)

Benefits

- Improved retention of complex concepts
- Improved skills
- Improved confidence

Concerns

- Reduced face-to-face
- User evaluation needed
- Possibility of training not user-friendly and/or efficient

Financial & Time Costs

- Initial costs
- Negative transfer
- Troubleshooting

► Product/Service Analysis (Cooper)

VIRTUAL REALITY

- ❖ Renders a digital environment completely separated from the real world
- ❖ Headsets are used to create a sense of immersion for the user

AUGMENTED REALITY

- ❖ Alters the users perception of the real world by displaying images and providing information
- ❖ Serves to elevate the users ability to observe and analyze their situation (Salehahmadi & Hajjaliasgari, 2019, 137)



► Marketing and Sales Strategies (Marisol)

Meta Partners With Children's Hospital Los Angeles

- Developed VR simulation for a low frequency, high-stakes procedure, status epilepticus
- Led to 11 more partnerships with other medical institutions
- Dr. Todd Change notes the future focus to create more trainings that fall within the parameters of low-frequency, high-stakes procedures



► Financial Analysis (Cooper)

Percent change between 2022 and 2023

Company		Revenue	Costs	Stock Value	Net Income (Operating Income for CAE)
	Meta	+16%	+1%	+195%	+69%
	Sony	+35%	+18%	+23.5%	+6%
	CAE Healthcare	+27%	+30%	+12%	+100%

Data retrieved from: Meta (2023, February 1), Meta(2023, May 31), Sony Group Corporation (2024), CAE Inc. (2023), Yahoo (Sony Group Corporation, 2024), Yahoo (CAE Inc, 2024)

► Recommendations (Cooper)



Ethics

- ❖ Medical practitioners are required to follow strict ethical standards
- ❖ Devices that cause harm to the user or patient will likely fail to receive FDA approval

Growth of the market

- ❖ Great potential as an investment opportunity
- ❖ Time is limited for other firms to enter the market and achieve success

Improvements

- ❖ Abatement of physical side effects
- ❖ Devices must be capable of adapting to the wide variety of medical situations and user needs
- ❖ Artificial Intelligence

► Conclusion (Cooper)

Key Concepts

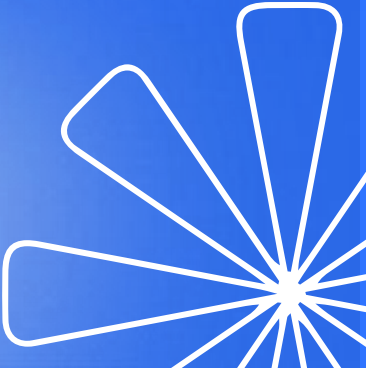
- ❖ Simulation has a long history within medical science, and extended reality is the future of this field
- ❖ Industry is growing rapidly and will continue to do so as long as the underlying technologies improve
- ❖ Meta is currently the dominant competitor
- ❖ The required complexity of the technology is the primary source of limitations and issues
- ❖ VR and AR devices have the potential to provide assistance across all aspects of healthcare and medical science





Click [here](#) to
view our
Reference
pages!

References





Thanks!

Do you have any questions?

amanda.m.clark@du.edu

cooper.sorich@du.edu

marisol.villalobos@du.edu

